RESEARCH REPORT

SEPTEMBER 2026

# Kubernetes as the Platform for AI Infrastructure

Adoption, core technologies, vendor offerings and production experience with training and inference on Kubernetes, 2025–2026


PUBLISHED
September 2026

EVIDENCE BASE
71 cited sources

PREPARED BY
Boku research agents


Research question: “How Kubernetes is being used for AI infrastructure”. Compiled from public sources by an automated multi-agent research pipeline. Claims are cited to their sources; numbered citations refer to the companion file kubernetes-platform-ai-infrastructure.references.json. Verify material facts before relying on them.

CONTENTS

# In this report

— Executive summary

— Key findings

01 **Why AI workloads are moving onto Kubernetes**

02 **Core building blocks: accelerator management with DRA**

03 **Training on Kubernetes: gang scheduling, operators and Slurm**

04 **Inference on Kubernetes: model-aware routing, disaggregated serving and autoscaling**

05 **Vendor and managed platform landscape**

06 **Production case studies**

07 **Limitations, operational challenges and open problems**

08 **Standards and community direction**

09 **Outlook and recommendations for platform teams**

— Conclusion

A Methodology and evidence quality

EXECUTIVE SUMMARY

# Kubernetes as the Platform for AI Infrastructure

As of September 2026, Kubernetes is the default control plane for AI inference. For training it is a growing platform, but it still depends on add-ons. In the CNCF 2025 Annual Cloud Native Survey, released in January 2026, 66% of organisations hosting generative AI models use Kubernetes for some or all of their inference, and 82% of container users run Kubernetes in production[1,2]. No comparable official figure exists for training on Kubernetes. Over half of organisations do not train models at all[1,3,4].

Adoption is broad but shallow. In the same survey, 44% of organisations do not yet run AI/ML workloads on Kubernetes, and only 7% deploy models daily[1,3]. The economics favour Kubernetes' strengths anyway. Gartner forecasts that inference will account for 55% ($23.3B) of AI-optimised IaaS spending in 2026, overtaking training for the first time, and 59% in 2027[5].

The inference stack has converged on shared upstream parts. These are the stable v1 InferencePool API with a model-aware Endpoint Picker[6,7,8], llm-d (a CNCF Sandbox project since March 2026)[9,10,11,12], and KServe's LLMInferenceService built on llm-d[13]. NVIDIA Dynamo with Grove is the main parallel stack[14,15]. KV-cache-aware routing is the best-evidenced performance lever. Google reports a 35% lower time to first token for one Vertex AI model and a 52% better P95 for another[16]. These figures, like all others in this area, are self-reported.

Accelerator management now rests on Dynamic Resource Allocation (DRA). Its core APIs went GA in v1.34 (September 2025)[17], and GPU partitioning, consumable capacity and device health moved to beta in v1.36 (April 2026)[18,19]. Training at scale is less mature. Native workload-aware (gang) scheduling is still alpha in v1.36[19,20], and GPU allocation in NVIDIA's DRA driver is experimental[21]. Large training clusters therefore rely on Kueue, Volcano, KAI Scheduler or Slurm running on Kubernetes[20,21,22,23,24,25].

Vendors build on the same primitives, checked through CNCF's Kubernetes AI Conformance programme. It launched in November 2025 and had 31 certified platforms by March 2026[26,27]. Vendors compete on control-plane scale, managed operations and vertical integration[16,24,25,26,28,29,30,31,32]. For platform teams, the practical priorities are the ones production users say matter most: queueing and GPU sharing to raise utilisation, and faster movement of model weights[33,34,35,36,37].

KEY FINDINGS

# What the evidence shows

01

### Kubernetes is mainly an inference platform for AI today.

In the CNCF 2025 survey (released 20 January 2026), 66% of organisations hosting generative AI models run some or all inference on Kubernetes[1,2]. No official training-on-Kubernetes figure exists, and over half of organisations do not train models[1,3,4].

02

### Adoption is wide but operationally immature.

44% of organisations do not yet run AI/ML on Kubernetes and only 7% deploy models daily[1,3]. A separate 2025 study found 35% of platform teams do not orchestrate AI workloads[38].

03

### DRA is the accelerator foundation, but gang scheduling for training is still alpha upstream.

Core DRA went GA in v1.34, and partitionable devices and device health went beta in v1.36[17,18,19]. Workload-aware scheduling and PodGroup are alpha in v1.36[19,20], so training clusters rely on Kueue, Volcano, KAI or Slurm-on-Kubernetes[20,21,22,23].

04

### The Kubernetes inference stack has converged on InferencePool, llm-d and LeaderWorkerSet, with NVIDIA Dynamo as the main alternative.

InferencePool is stable at v1 and its production Endpoint Picker now lives in llm-d[6,7,8]. KServe v0.17 and OpenShift AI 3 build on llm-d[13,32]. Grove is being integrated with llm-d[31,39].

05

### KV-cache-aware routing delivers the best-documented gains, all of them self-reported.

Google reports up to 40% lower TTFT and inter-token latency (ITL) from llm-d predicted-latency scheduling in Vertex AI[40]. Baseten reports 48–49% lower P95/P99 latency with Dynamo's router[41]. Tesla reports 3x output tokens/s on one deployment[42,43].

06

### Utilisation and cost gains come from scheduling and sharing more than from hardware.

China Merchants Bank pooled nearly 10,000 accelerators under Kueue, KEDA, HAMi and Fluid[33]. It raised utilisation from 35% to over 60% and cut inference cost per million tokens by over 60%[33].

07

**At the frontier, power and networking limit scale more than Kubernetes does.**

Google ran a 130,000-node experimental GKE cluster and named electrical power as the main constraint (about 2,700 W per GB200 GPU)[28]. EKS has supported 100,000 nodes since July 2025[29].

01 SECTION

# Why AI workloads are moving onto Kubernetes

The strongest adoption signal is about inference. The CNCF 2025 Annual Cloud Native Survey was released on 20 January 2026. It found that 66% of organisations hosting generative AI models use Kubernetes for some or all of their inference workloads, and that 82% of container users run Kubernetes in production, up from 66% in 2023[1,2]. Official materials give only the combined 66% figure. We could not confirm a reported split into 'all' versus 'some' inference in any source[1,3,4,44].

Beneath that headline, adoption is shallow. Only 7% of organisations deploy AI models daily, 47% deploy them occasionally, and 44% do not yet run AI/ML workloads on Kubernetes at all[1,3]. A CNCF blog from August 2026 cites 2025 platform-engineering research finding that 35% of platform teams still do not orchestrate AI workloads[38].

TABLE 1

**Key figures from the CNCF 2025 Annual Cloud Native Survey**

| METRIC | VALUE | NOTE |
|---|---|---|
| **Container users running Kubernetes in production** | 82% | Up from 66% in 2023 |
| **GenAI-hosting organisations using Kubernetes for some or all inference** | 66% | No official all/some split |
| **Organisations not yet running AI/ML on Kubernetes** | 44% | Official announcement |
| **Organisations deploying AI models daily** | 7% | 47% deploy occasionally |
| **Organisations that do not train models** | Over half | Linux Foundation blog |

Source: [1], [2], [3], [4], [44]. Data as of 2025 survey, published 2026-01-20. Secondary coverage reports a sample of 628 respondents. The official CNCF and Linux Foundation pages do not state the sample size[1,4,45,46].

NO TRAINING-ADOPTION FIGURE

No official CNCF or Linux Foundation page gives a share of organisations training models on Kubernetes. The 36% 'training' figure in the press release is a skills challenge, not model training[1,3,4]. Public survey data covers inference far better than training[1,3,47].

## Spending is shifting to inference

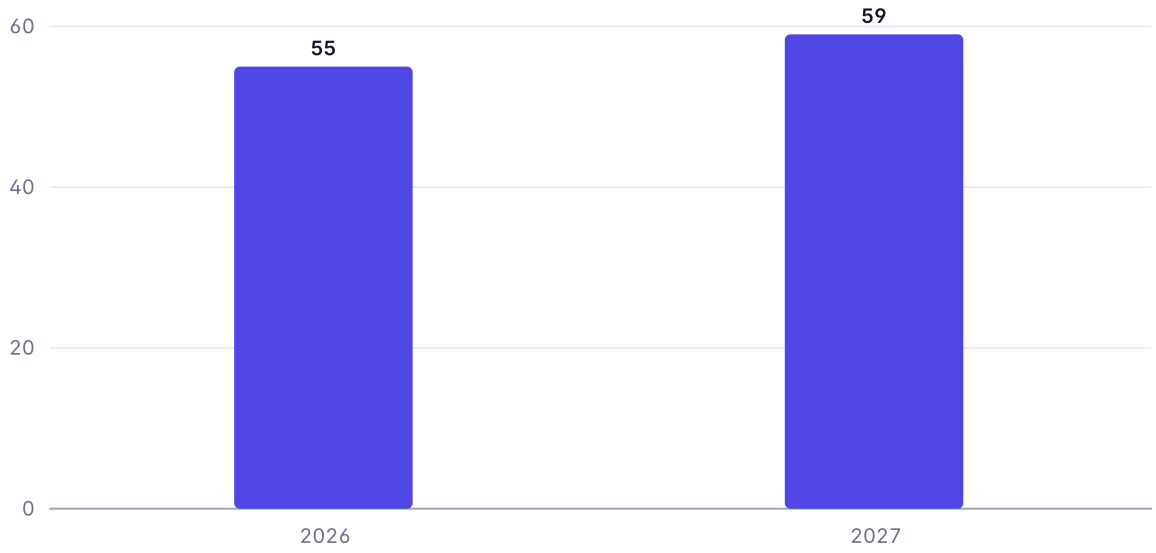
Gartner (August 2026) forecasts worldwide AI-optimised IaaS spending to grow 96% to about $42B in 2026. Of that, $23.3B (55%) is forecast for inference and $19B for training. Gartner expects the inference share to reach 59% in 2027, in a market of about $66B[5]. Deloitte

measures compute rather than spending. It estimates inference at about one-third of AI compute in 2023 and about half in 2025, and projects about two-thirds in 2026[48].

FIGURE 1

**Inference share of AI-optimised IaaS spending (Gartner forecast)**

% of spending




Source: [5]. Data as of Forecast published 2026-08-10. Estimates, not measured spending.

The two forecasts measure different things and neither is a precise market share, but they point the same way. This suggests that the elastic, service-style workloads Kubernetes handles best will make up a growing share of AI infrastructure[5,48].

02 SECTION

# Core building blocks: accelerator management with DRA

Dynamic Resource Allocation replaces the fixed-count device-plugin model with claimable, parameterised devices. Its core resource.k8s.io APIs reached GA in Kubernetes v1.34 on 1 September 2025[17]. v1.36 (22 April 2026) moved the features that matter most for GPUs into beta, enabled by default. These are partitionable devices, consumable capacity, and device taints and tolerations[18,19].

TABLE 2

**DRA and related feature maturity by Kubernetes release**

| FEATURE | STATUS | RELEASE | WHAT IT ENABLES |
|---|---|---|---|
| **Core DRA APIs (resource.k8s.io)** | GA | v1.34 (Sep 2025) | Structured device claims beyond device plugins |
| **AdminAccess for ResourceClaims (KEP-5018)** | GA | v1.36 (Apr 2026) | Privileged admin access to in-use devices |
| **Partitionable devices** | Beta, on by default | v1.36 | GPU slicing (e.g. MIG-style partitions) |
| **Consumable capacity** | Beta, on by default | v1.36 | Finer-grained device sharing |
| **Device taints and tolerations** | Beta | v1.36 | Keep specialised devices for the right workloads |
| **ResourceClaim device status** | Beta | v1.36 | Device status reporting on claims |
| **allocatedResourcesStatus in Pod status (KEP-4680)** | Beta | v1.36 | Shows Unhealthy/Unknown devices behind crash loops |
| **ResourceClaims for PodGroups (KEP-5729)** | Alpha in v1.36; beta in v1.37 | v1.36 / v1.37 | Claim reserved per group; removes the 256-pod reservedFor limit |

Source: [17], [18], [19], [20], [49], [50]. Data as of 2026-05.

Device health reporting closes a long-standing operational blind spot. With allocatedResourcesStatus at beta, operators can see in 'kubectl describe pod' whether a crash loop comes from an unhealthy device. This works for both device-plugin and DRA devices[19].

## NVIDIA's DRA driver and multi-node NVLink

NVIDIA's DRA driver provides ComputeDomains, an officially supported abstraction for secure Multi-Node NVLink on systems such as GB200. It sets up IMEX channels and ties the domain's lifetime to the workload[21]. The driver's GPU-allocation component is still experimental and off by default[21]. At KubeCon Europe in March 2026, NVIDIA donated the driver to community governance under the Kubernetes project[31,39].

> **CAUTION**
>
> DRA is GA as an API, but the most common use, allocating NVIDIA GPUs through DRA, is not yet officially supported in NVIDIA's driver[21]. Most GPU clusters still depend on device plugins or vendor tooling for day-to-day allocation. This last point is our inference.

WG Device Management set its 2026 priorities in June. They are multi-node allocation through integration with workload-aware scheduling, device health and failure detection, advanced sharing, and portable specifications[51].

03 SECTION

# Training on Kubernetes: gang scheduling, operators and Slurm

Distributed training needs all-or-nothing placement: a job whose workers cannot all start wastes every GPU it holds. Kubernetes first added native gang scheduling as alpha in v1.35[20,50,52,53]. v1.36 reworked it into Workload Aware Scheduling, with a static Workload API, a decoupled runtime PodGroup API integrated with the Job controller, and a scheduling cycle that evaluates each group atomically. All of it is still alpha[19,20].

Production training therefore runs on add-on schedulers. Kueue's queueing, quotas with fair sharing and preemption, timeout-based all-or-nothing admission and topology-aware scheduling are GA. MultiKueue multi-cluster dispatch reached GA in February 2026[54,55]. Kueue integrates with Job, JobSet, Kubeflow jobs, Ray and LeaderWorkerSet, and its latest stable release is v0.19.5 from 17 September 2026[54,55]. Volcano v1.14 added gang scheduling at PodGroup and SubGroup level with network-topology-aware bin-packing. v1.15 (30 May 2026) added gang-level preemption and brought DRA devices under queue quotas[56,57].

NVIDIA's KAI Scheduler, the open-sourced Run:ai scheduler, joined the CNCF Sandbox in March 2026[31,39]. It provides the hierarchical gang scheduling that Grove relies on[14,15]. On the operator side, Kubeflow Trainer v2.2 (20 March 2026) merges the old per-framework job types into one TrainJob API on JobSet. It lists DRA, multi-node NVLink and native workload-aware scheduling as roadmap items[23]. Managed platforms fill the same gap. SageMaker HyperPod on EKS added gang scheduling with timeout-based requeueing in April 2026[58].

TABLE 3

**Training schedulers and Slurm-on-Kubernetes options**

| OPTION | GANG SCHEDULING | TOPOLOGY AWARENESS | MULTI-CLUSTER | MATURITY SIGNAL |
|---|---|---|---|---|
| **Kubernetes WAS / PodGroup** | Native, atomic group cycle | Included in WAS work | No | Alpha in v1.36 |
| **Kueue** | Timeout-based all-or-nothing admission (GA) | Topology-aware scheduling (GA) | MultiKueue (GA Feb 2026) | v0.19.5, Sep 2026 |
| **Volcano** | PodGroup and SubGroup gangs; gang-level preemption | HyperNode (switch/rack) bin-packing | Not documented in corpus | v1.15, May 2026 |
| **KAI Scheduler** | Hierarchical gang scheduling (used by Grove) | Topology-aware placement | Not documented in corpus | CNCF Sandbox, Mar 2026 |
| **Slurm on Kubernetes (SUNK, Slinky)** | Slurm semantics | Topology-aware scheduling (SUNK); dynamic topology (Slinky v1.1.0) | Not documented in corpus | Production at CoreWeave and NVIDIA |

Source: [14], [15], [19], [20], [24], [25], [31], [39], [54], [55], [56], [57]. Data as of 2026-09.

## Slurm and Kubernetes are converging

CoreWeave's SUNK runs Slurm login and compute nodes as pods on its Kubernetes service. A syncer and a Slurm-aware pod scheduler let batch jobs and Kubernetes services share one compute pool, with GPU straggler detection added[24]. NVIDIA reports running SchedMD's Slinky slurm-operator across internal clusters totalling more than 8,000 GPUs. It measured NCCL collective performance matching non-containerised Slurm, with a limit of one worker pod per node[25]. NVIDIA announced its acquisition of SchedMD on 15 December 2025 and committed to keeping Slurm open source[59].

> **INSIGHT**
>
> Teams with HPC-style training habits do not have to choose between Slurm and Kubernetes. The evidence points to Slurm semantics running on a Kubernetes substrate that is shared with inference[24,25]. This is our interpretation.

04 SECTION

# Inference on Kubernetes: model-aware routing, disaggregated serving and autoscaling

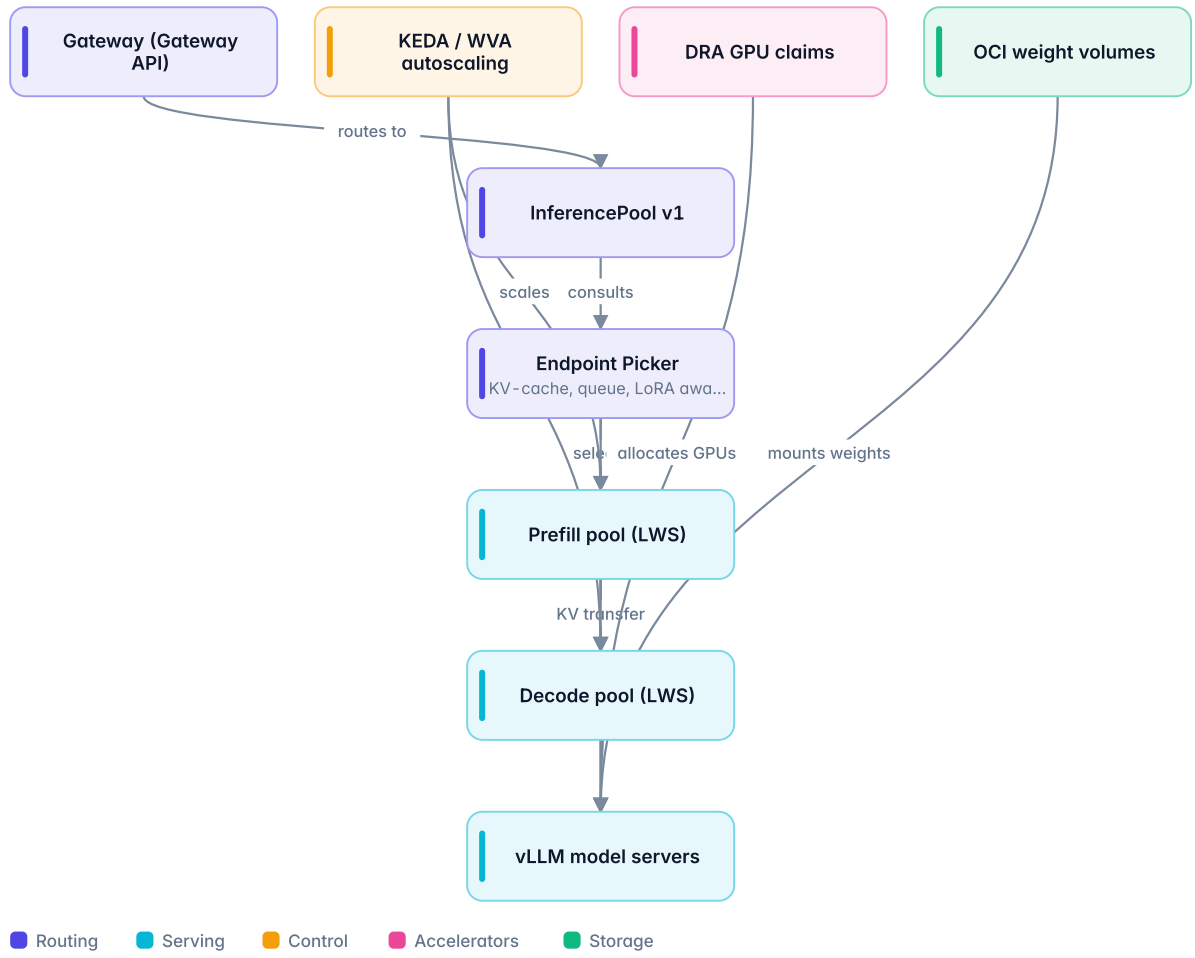
The Gateway API Inference Extension is the routing layer. Its InferencePool API is stable at v1 and groups model-server Pods behind a standard Gateway such as Envoy Gateway, kgateway, GKE Gateway, NGINX or Istio. An Endpoint Picker (EPP) chooses a replica for each request from its KV-cache utilisation, queue length and active LoRA adapters[6,7,8]. In v1.6.0 the production EPP moved into the llm-d repository, and a lightweight reference EPP remains for conformance[6,7,8].

llm-d supplies the serving stack: vLLM plus the inference scheduler, tiered KV-cache offload, prefill/decode disaggregation and wide expert parallelism. It joined the CNCF Sandbox in March 2026[9,10,11,12]. Its v0.9 release on 28 August 2026 added router high availability, KEDA autoscaling on KV-cache pressure and queue depth, and revision-aware routing for disaggregated deployments[9,10,11,12]. The same release added support for GB200/GB300, Intel XPU, ROCm and TPU v7[9,10,11,12].

Higher-level platforms wrap these parts. KServe v0.17 (13 March 2026) made LLMInferenceService production-ready on llm-d. It adds KV-cache-aware routing via the EPP, prefill/decode disaggregation, and multi-node tensor, data and expert parallelism via LeaderWorkerSet. Its Workload Variant Autoscaler scales on KV-cache use and queue depth[13]. LeaderWorkerSet deploys a leader and its workers as one replica. Its DisaggregatedSet API, co-designed with llm-d, coordinates prefill/decode rollouts, though its own gang scheduling is still alpha[22].

FIGURE 2

**Converged Kubernetes LLM serving stack**




Source: [6], [7], [8], [9], [10], [11], [12], [13], [18], [19], [22].

## NVIDIA Dynamo and Grove

NVIDIA Dynamo models an inference graph as a custom resource managed by an operator. Grove orchestrates the router, prefill and decode roles as PodCliques inside a PodCliqueSet, and uses KAI Scheduler for hierarchical gang scheduling[14,15]. Dynamo 1.0 shipped on 16 March 2026 with 17 named production adopters and integrations with managed services including EKS and AKS[60]. NVIDIA says Grove's API is being integrated with llm-d, which should narrow the gap between the two stacks[31,39].

## Model weights and cold starts

Getting weights onto GPUs quickly is the main cold-start problem. OCI Artifact and Image volume sources (KEP-4639) went stable in v1.36, so weights can be packaged as OCI artifacts and mounted from a registry[19]. On GKE, the Cloud Storage FUSE serving profile cut load time for a 480 GB Qwen3-235B-A22B model on TPUs from 39 hours to 14 minutes, according to Google (April 2026)[61].

**CAUTION**

No independent, like-for-like benchmark compares llm-d, KServe, Dynamo and other stacks. Every performance figure in this report comes from a vendor or adopter.

05 SECTION

# Vendor and managed platform landscape

The major platforms package the same upstream primitives and differentiate on three things: control-plane scale, managed operations and vertical integration[16,24,25,26,28,29,30,31,32]. The first certified platforms in CNCF's AI Conformance programme (November 2025) were EKS, GKE, Microsoft Azure, Red Hat OpenShift and Broadcom VKS, with CoreWeave, Oracle Cloud and Akamai among the supporters[26].

TABLE 4

**How major vendors package AI on Kubernetes**

| VENDOR | CLUSTER SCALE | INFERENCE LAYER | TRAINING / SCHEDULING LAYER | NOTABLE DIFFERENTIATOR |
|---|---|---|---|---|
| **Google GKE** | 65,000 nodes supported; 130,000 in an experiment (Nov 2025) | GKE Inference Gateway (prefix-cache and load-aware) | Kueue; TPU dynamic slicing | Spanner-based control-plane store; GCS FUSE serving profile |
| **Amazon EKS / SageMaker HyperPod** | 100,000 nodes (since Jul 2025) | Dynamo integration; gateway details not researched | HyperPod task governance with gang scheduling (Apr 2026) | Reworked etcd storage layer |
| **Microsoft AKS** | Not stated in corpus | KAITO (vLLM), AI Runway, Dynamo-Grove on AKS | AKS on Bare Metal (preview), Anyscale managed Ray (preview) | Fleet Manager GA for Arc-enabled clusters |
| **Red Hat OpenShift AI 3** | Not stated in corpus | llm-d GA (NVIDIA and AMD) | Not covered in corpus | Model-as-a-Service, AI Hub, agent API |
| **NVIDIA** | N/A (software) | Dynamo 1.0, Grove | KAI Scheduler, DRA driver, Slurm via SchedMD/Slinky | Vertical integration across the stack |
| **CoreWeave** | Not stated in corpus | llm-d founding contributor | SUNK (Slurm on CKS) | Slurm and Kubernetes share one compute pool |

Source: [9], [10], [11], [12], [14], [15], [16], [24], [25], [28], [29], [30], [31], [32], [39], [58], [59], [60], [61], [62]. Data as of 2026-09.

Scale engineering is where the hyperscalers diverge most visibly. Google's 130,000-node experiment replaced etcd with a Spanner-based store and handled 13,000 QPS of lease updates. It created 130,000 Pods in 3 minutes 40 seconds and preempted 39,000 Pods in 93 seconds[28]. AWS rebuilt EKS's etcd storage layer to support 100,000 nodes, which it equates to up to 800,000 NVIDIA GPUs[29].

Microsoft's Build 2026 announcements follow a strategy of wrapping open source (Kubernetes, Ray, Gateway API) in managed services[30]. Red Hat made llm-d generally available in OpenShift AI 3 in October 2025[32]. NVIDIA is the most vertically integrated. It spans GPU allocation (DRA

driver), scheduling (KAI), serving (Dynamo, Grove) and, through SchedMD, Slurm[31,39,59].

> **NOTE**
>
> The corpus contains no cost comparison between managed offerings, and no detail on the EKS and AKS inference gateways. Treat the table as a capability map, not a buyer's ranking.

06 SECTION

# Production case studies

## China Merchants Bank: one pool for training and inference

Context and problem: training needs stable reserved capacity while inference needs fast elastic scaling. China Merchants Bank had nearly 10,000 heterogeneous accelerator cards to serve both[33].

Approach: the bank built one Kubernetes control plane. It uses Kueue for admission and quotas, KEDA and Prometheus for inference autoscaling, HAMi for fine-grained accelerator sharing and Fluid for faster access to data and weights, with 99% of accelerator resources under the unified framework[33].

Results: average accelerator utilisation rose from 35% to over 60%, and inference cost per million tokens fell by more than 60%[33]. Five LoRA fine-tuning tenants sharing one base-model instance cut accelerator use by 80%[33].

Lessons: in this case the gains came from queueing, sharing and pooling, not from new hardware. This is consistent with other deployments, where default Kubernetes scheduling proved insufficient for shared GPUs[33,34,35,36,37].

## Other named deployments

TABLE 5

**Named AI-on-Kubernetes deployments and reported outcomes**

| ORGANISATION | WORKLOAD | STACK | REPORTED OUTCOME | SOURCE / DATE |
|---|---|---|---|---|
| **Google Vertex AI** | LLM inference | GKE Inference Gateway; llm-d predicted-latency scheduling | TTFT −35% (Qwen3-Coder); P95 TTFT −52% (DeepSeek V3.1); up to 40% lower TTFT/ITL | Google, Feb and Mar 2026 |
| **Tesla** | Hundreds of LLMs | KServe LLMInferenceService, llm-d, vLLM, Envoy, Inference Extension | 3x output tokens/s, 2x lower TTFT on one deployment (Llama 3.1 70B, 4x MI300X) | KServe/llm-d blog, Apr 2026 |
| **Baseten** | LLM inference (Qwen3 Coder 480B) | NVIDIA Dynamo KV-aware router | P95 −48%, P99 −49% latency on production traffic | Baseten, Mar 2026 |
| **Wayve** | Multi-tenant inference, evaluation, synthetic data | Kueue, Ray | Predictable allocations and better utilisation (qualitative) | KubeCon EU keynote, Mar 2026 |
| **Subaru** | Driver-assistance AI development | On-prem Kubernetes, Gateway API, Harbor, Argo | 30 GB+ image pulls from ~3 hours to ~3 minutes | CNCF, Jul 2026 |

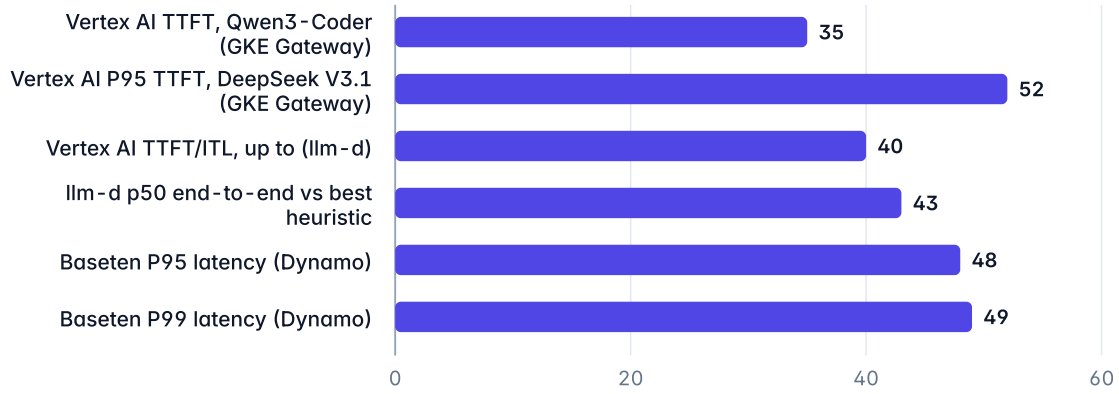
| ORGANISATION | WORKLOAD | STACK | REPORTED OUTCOME | SOURCE / DATE |
|---|---|---|---|---|
| **Anthropic, Amazon Nova** | Ultra-scale workloads on EKS | EKS 100K-node clusters | Anthropic: writes within 15 ms rose from 35% to 90% | AWS, Jul 2025 (historical) |
| **Uber** | Michelangelo ML training | Ray on Kubernetes, federated scheduler | 1.5–4x faster training | Uber, Apr 2025 (historical) |

Source: [16], [29], [34], [35], [37], [40], [41], [42], [43]. Data as of 2026-09.

FIGURE 3

**Self-reported latency reductions from model-aware routing**

% reduction




Source: [16], [40], [41]. Data as of 2026-02 to 2026-03. Different models, workloads and baselines. Not directly comparable.

**WEAKER-SOURCED FIGURES**

HAMi's KubeCon China recap reports that ICBC's GPU utilisation rose from 20% to 70% and that SF Express cut its GPUs from 1,400 to 1,000. These are self-reported in a project recap[63]. The Character.AI figures (60% faster model loading, 50% lower TCO) and Anthropic's TPU dynamic-slicing figures trace only to conference videos[36,62,64,65,66]. Google's official documentation states dynamic slicing gives up to 5x faster startup and 4.5x faster recovery[62].

Earlier milestones are historical. OpenAI reported scaling one Kubernetes cluster to 7,500 nodes in January 2021, per a secondary summary[67]. In November 2024, Anthropic's Head of Compute endorsed GKE's 65,000-node support as 'the scale we need'[68]. That quote does not by itself establish which workloads Anthropic runs on GKE.

07 SECTION

# Limitations, operational challenges and open problems

- Gang scheduling is not native yet. Workload Aware Scheduling and PodGroup are alpha in v1.36, and LeaderWorkerSet gang scheduling is alpha[19,20,22]. Training clusters depend on Kueue, Volcano, KAI or managed equivalents[20,21,22,23].
- GPU allocation through DRA is not production-ready from NVIDIA. The driver's GPU-allocation path is experimental and off by default, and dynamic MIG is still planned[21]. Kubeflow Trainer lists DRA and multi-node NVLink as future work[23].
- Utilisation and multi-tenancy are the main cost levers. Default scheduling falls short for shared GPUs[35], and China Merchants Bank started from 35% average utilisation[33]. The corpus has no tier-1 comparison of the isolation trade-offs between hardware partitioning and software sharing such as HAMi.
- Moving weights and images is a bottleneck. Tesla cited storage problems with multi-hundred-GB models and manual recovery from local-PV affinity[42,43]. Subaru's 30 GB+ images took about 3 hours to pull before its platform rebuild[34].
- At the frontier, the limits are power, networking and multi-cluster operation. Google names electrical power as the main limit on scale and lists managed DRANET for RDMA and MultiKueue as future work[28].
- Evidence on failure handling and goodput is thin. Beyond NVIDIA's finding that Slinky adds no measurable NCCL overhead[25], we found no quantitative comparison of large-run resilience on Kubernetes versus bare-metal Slurm.

> NOTE
>
> Non-NVIDIA accelerators are under-documented. Apart from release notes, such as llm-d v0.9 support for ROCm, Intel XPU and TPU v7[9,10,11,12], the corpus has one AMD production data point (Tesla on MI300X)[42,43].

08 SECTION

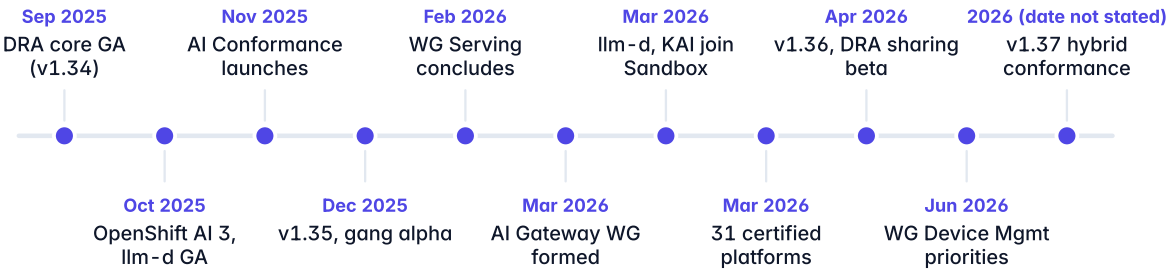
# Standards and community direction

CNCF's Certified Kubernetes AI Conformance programme launched on 11 November 2025. It covers GPU resource management, distributed scheduling, accelerator autoscaling and monitoring[26]. By 24 March 2026, 31 platforms were certified, up from 18. The v1.35 requirements added high-performance pod-to-pod communication, advanced inference ingress, disaggregated inference and agentic workloads[27]. The conformance repository accepts certifications for v1.33 through v1.37, each valid for one year. v1.37 introduced hybrid verification that combines automated tests with manual attestation[69].

Community governance has spread out from a single working group. WG Serving concluded on 26 February 2026 and handed its work to SIG Node, Scheduling, Apps, Network and Scalability and to WG Device Management[70]. A new AI Gateway Working Group, announced on 9 March 2026, builds on the Inference Extension to standardise networking for AI traffic[71].

FIGURE 4

**Key milestones for AI on Kubernetes**




Source: [9], [10], [11], [12], [17], [18], [19], [20], [26], [27], [31], [32], [39], [50], [51], [52], [53], [69], [70], [71].

09 SECTION

# Outlook and recommendations for platform teams

The recommendations below are our interpretation of the evidence, ordered by how mature the underlying components are.

- Adopt Kueue now for queueing, quotas and all-or-nothing admission. Its core features and MultiKueue are GA[54,55], and it underpins the China Merchants Bank, Wayve and Google 130,000-node results[28,33,35].
- Plan device management around DRA. Use the v1.36 beta sharing and health features[18,19], and keep device plugins or vendor tooling for NVIDIA GPU allocation until the driver's path leaves experimental status[21].
- Standardise inference routing on InferencePool with a production EPP. Choose llm-d/KServe or Dynamo for the serving layer, and design autoscaling around KV-cache and queue signals rather than CPU[6,7,8,9,10,11,12,13].
- Treat native gang scheduling as upcoming, not available. Workload Aware Scheduling is alpha[19,20], and Kueue and Volcano API versions are still changing (Kueue v0.20 stops serving v1beta1)[54,55].
- Budget for weight distribution. Evaluate OCI image volumes, now stable[19], and storage-side acceleration such as GKE's FUSE serving profile[61].
- Require current AI Conformance certification from vendors, and check the Kubernetes version it covers[69].

INSIGHT

The largest reported gains come from pooling and sharing GPUs across tenants and workload types, not from swapping serving frameworks[33,34,35,36,37]. Organisational and scheduling work is likely to repay more than stack selection. This is our inference.

CONCLUSION

# Conclusion

For most organisations, the question is no longer whether to run AI inference on Kubernetes. It is how quickly they can move from occasional deployments to operating AI as a routine platform service. The 7% of organisations deploying models daily[1,3] marks how early most teams still are. The forecast shift of spending toward inference[5] means that gap will increasingly show up as cost.

Three developments will decide whether Kubernetes also becomes the default for large-scale training. The first is promotion of Workload Aware Scheduling and PodGroup-scoped DRA claims beyond alpha and beta[19,20,49,50]. The second is official GPU allocation in NVIDIA's community-governed DRA driver[21,31,39]. The third is whether Slurm-on-Kubernetes under NVIDIA's ownership of SchedMD stays vendor-neutral in practice[25,59].

On inference, watch whether Grove's integration with llm-d merges the two leading stacks[31,39], and whether the AI Gateway Working Group standardises traffic handling beyond single-model routing[71]. The evidence also still lacks independent benchmarks. Until neutral, like-for-like results exist, platform teams should reproduce vendor latency claims on their own traffic before committing.

---

Numbered citations refer to 71 sources listed in the companion file **kubernetes-platform-ai-infrastructure.references.json**. Re-render with `--save-ref` to include them here.

APPENDIX A

# Methodology and evidence quality

This report was produced by Boku, a multi-agent research pipeline. Research workstreams (technical, market, competitive, case-study) gathered evidence from public sources; each claim was recorded with its sources and date. An independent fact-checking agent reviewed the corpus over 2 round(s) (final status: pass) before synthesis and editing. Information dated within 365 days of publication is labelled current; older material is labelled historical.

TABLE 6

### Evidence base

| MEASURE | COUNT |
|---|---|
| **Findings recorded** | 72 |
| **Findings used (after fact-check)** | 70 |
| **Findings rejected** | 2 |
| **Findings flagged as weakly supported** | 8 |
| **Distinct sources supporting usable findings** | 75 |
| **Tier 1 · Primary / official** | 56 |
| **Tier 2 · Authoritative secondary** | 7 |
| **Tier 3 · Specialist / analyst** | 9 |
| **Tier 4 · Community / unverified** | 3 |

## Limitations and unresolved questions

- Unverified: F055 p50 TTFT 0.97s / 70% reduction figures in the llm-d predicted-latency blog
- Unverified: F058 Baseten P95/P99 reductions
- Unverified: F024 v1.37 hybrid verification in the k8s-ai-conformance README
- No official figure for the share of organisations running AI training (as opposed to inference) on Kubernetes. The official survey sample size is not stated (628 comes from secondary coverage only).
- No independent, like-for-like benchmark comparing llm-d, KServe, Dynamo and other serving stacks. All performance figures are self-reported by vendors or adopters.
- Thin evidence on multi-node networking (RDMA/InfiniBand, DRANET) in production on Kubernetes beyond Google's mention of managed DRANET as future work.

- Little quantitative data on failure handling and job goodput for large training runs on Kubernetes compared with bare-metal Slurm, beyond NVIDIA's statement that Slinky adds no NCCL overhead.
- Multi-tenancy security and isolation for shared GPUs (e.g. MIG/MPS vs software sharing such as HAMi) is not covered by tier-1 comparative evidence.
- No cost comparison between managed Kubernetes AI offerings, and no per-vendor details on the EKS and AKS inference gateways.